Purabh Singh

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Education

Rutgers University Aug 2023 – May 2027
B.S. in Computer Science, Minor in Data Science *New Brunswick, NJ*

Technologies

Languages: Python, Java, TypeScript, SQL, C++, JavaScript, HTML **Frontend**: React, Angular, Swift, Next.js, AJAX **Backend**: Node.js, NestJS, LangChain, Express, Spring Boot, Pandas, Docker, Jira **Blockchain**: Solidity, Hyperledger Besu, Web3j, ethers.js, Web3.js **Database**: PostgreSQL, MySQL, SQLite, GitHub **Networking/Protocols**: REST, WebSockets **Operating Systems**: Windows, Linux

Achievements

- Winner – MusaLabs Hackathon Sponsored by Microsoft Azure (1st of 50+ teams)
- Dean's List

Experience

Software Engineering Intern Summer 2026
BNY *Pittsburgh, PA*
- Member of BNY's mission-critical funds-control engine team, a Java 21 / Spring Boot 3.5 platform processing billions in daily payment volume across 5 concurrent JVM lanes; a single minute of downtime carries an estimated $3.2B exposure.
- Built an immutable audit trail using Hyperledger Besu, Solidity smart contracts, and Web3j, writing every cleared debit and credit to a private, permissioned blockchain, enabling real-time reconciliation across 3 fragmented systems with zero impact to payment latency.
- Developed an Angular front-end powered by ethers.js that reads directly from the Besu network, cutting network hops from 2 to 1 and delivering live balance and ledger visibility even during backend downtime.

Software Engineering Intern May 2025 – Aug 2025
Essex Securities *Jersey City, NJ*
- Designed and implemented a LangChain-powered AI tool that automated the extraction and comparison of actual vs estimated EPS from quarterly earnings reports, reducing manual review time by over 70%. Built a retrieval-augmented generation pipeline with LLMs, pgvector semantic search, and FastAPI endpoints, with extensions planned for SEC filings and earnings call transcripts.
- Built an MVP crypto exchange entirely in Python using FastAPI, PostgreSQL, and asyncio, enabling secure, low-latency trading for a single spot market. Developed an in-memory matching engine, REST/WebSocket APIs, and a double-entry ledger to ensure accurate order execution and balance reconciliation. Streamlined operations by integrating risk controls, basic KYC, and automated trade settlement into a unified backend service.

Software Engineering Intern Jul 2024 – Oct 2024
Essex Securities *Jersey City, NJ*
- Automated limit order execution through Bloomberg EMSX API integration, streamlining workflows and reducing manual entry time by 20%.
- Revamped brokerage portal with a modern UI and faster navigation. Developed a job board using Dropbox for job descriptions and Google Forms for applications, with a dynamic, filterable front-end and embedded JD previews to streamline hiring.

Software Engineering Intern Jun 2024 – Jul 2024
Azonix Tech Private Ltd *Chennai, India*
- Resolved user experience issues by identifying key flaws and deploying UI improvements to production.
- Streamlined EV charger data analysis by designing a dashboard to visualize performance metrics.
- Implemented facial recognition on CCTV feeds, enabling 90%+ accuracy in varied conditions.

Projects

LeaseLink June 2025
- Won the MusaLabs Hackathon out of 50+ teams and 150+ participants with a Tinder-style housing marketplace that connects renters and landlords through mutual interest.
- Architected and developed a full-stack platform with a React + Tailwind frontend, Express.js backend, PostgreSQL database, and Google OAuth for secure authentication.
- Led backend development of a gated real-time chat system via WebSockets and implemented landlord approval workflows, enabling streamlined property discovery and efficient communication.

Adventure Catalyst Apr 2024 – Aug 2024
- Created and balanced a custom Minecraft modpack using Java and introducing new mechanics and creative elements.
- Addressed gameplay imbalances by modifying maps, adding new block types, and enhancing the player experience.